



Distribution of vasopressin 1a and oxytocin receptor protein and mRNA in the basal forebrain and midbrain of the spiny mouse (*Acomys cahirinus*)

Jeanne M. Powell¹ · Kiyoshi Inoue^{2,3} · Kelly J. Wallace¹ · Ashley W. Seifert⁴ · Larry J. Young^{2,3,5} · Aubrey M. Kelly¹

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Abstract

The nonapeptide system modulates numerous social behaviors through oxytocin and vasopressin activation of the oxytocin receptor (OXTR) and vasopressin receptor (AVPR1A) in the brain. OXTRs and AVPR1As are widely distributed throughout the brain and binding densities exhibit substantial variation within and across species. Although OXTR and AVPR1A binding distributions have been mapped for several rodents, this system has yet to be characterized in the spiny mouse (*Acomys cahirinus*). Here we conducted receptor autoradiography and in situ hybridization to map distributions of OXTR and AVPR1A binding and *Oxtr* and *Avpr1a* mRNA expression throughout the basal forebrain and midbrain of male and female spiny mice. We found that nonapeptide receptor mRNA is diffuse throughout the forebrain and midbrain and does not always align with OXTR and AVPR1A binding. Analyses of sex differences in brain regions involved in social behavior and reward revealed that males exhibit higher OXTR binding densities in the lateral septum, bed nucleus of the stria terminalis, and anterior hypothalamus. However, no association with gonadal sex was observed for AVPR1A binding. Hierarchical clustering analysis further revealed that co-expression patterns of OXTR and AVPR1A binding across brain regions involved in social behavior and reward differ between males and females. These findings provide mapping distributions and sex differences in nonapeptide receptors in spiny mice. Spiny mice are an excellent organism for studying grouping behaviors such as cooperation and prosociality, and the nonapeptide receptor mapping here can inform the study of nonapeptide-mediated behavior in a highly social, large group-living rodent.

Keywords Vasopressin · Oxytocin · Receptor autoradiography · In situ hybridization · Social behavior · Spiny mice

Abbreviations

AOB Accessory olfactory bulb
 aco Anterior commissure
 AI Agranular insular area of the cortex
 AHAL Amygdalohippocampal area, anterolateral part

AH Anterior hypothalamus
 AOA Anterior olfactory areas
 APT Anterior pretectal nucleus
 BLA Basolateral amygdala
 BNST Bed nucleus of the stria terminalis
 BNSTa Bed nucleus of the stria terminalis, anterior division
 BNSTp Bed nucleus of the stria terminalis, posterior division
 CPu Caudate putamen
 CeA Central amygdala
 Cg Cingulate cortex
 cc Corpus callosum
 Cg2 Cortical layer 2
 Cg5 Cortical layer 5
 DG Dentate gyrus
 VDB Diagonal band nucleus
 DI Dysgranular insular cortex
 GP Globus pallidus

✉ Aubrey M. Kelly
aubrey.kelly@emory.edu

¹ Department of Psychology, Emory University, 36 Eagle Row, Atlanta, GA 30322, USA

² Center for Translational Social Neuroscience, Emory University, Atlanta, GA 30329, USA

³ Emory National Primate Research Center, Emory University, Atlanta, GA 30329, USA

⁴ Department of Biology, University of Kentucky, 101 Morgan Building, Lexington, KY 40506, USA

⁵ Department of Psychiatry and Behavioral Sciences, Emory University School of Medicine, Atlanta, GA 30322, USA

Hi	Hippocampus
CA1	Hippocampal field CA1
CA1py	Hippocampal field CA1 pyramidal layer
CA3	Hippocampal field CA3
IP	Interpeduncular nucleus
LD	Lateral dorsal nucleus of thalamus
LH	Lateral hypothalamus
LS	Lateral septum
LS _c	Lateral septum, caudo-dorsal region
LS _v	Lateral septum, ventral region
MOB	Main olfactory bulb
MOB _{gl}	Main olfactory bulb, glomerular layer
MOB _{gr}	Main olfactory bulb, granule layer
MeA	Medial amygdala
MG	Medial geniculate nucleus of the thalamus
MM	Medial mammillary nucleus of the hypothalamus
MPOA	Medial preoptic area of the hypothalamus
NAcc	Nucleus accumbens
OT	Olfactory tubercle
och	Optic chiasm
PV	Paraventricular thalamic nucleus
PVA	Paraventricular thalamic nucleus, anterior division
PAG	Periaqueductal gray
PVG	Periventricular gray
PIR	Piriform area
Re	Reuniens thalamic nucleus
SC	Superior colliculus
SuG	Superior colliculus, superficial gray, sensory related
SC _m	Superior colliculus, motor related
SCN	Suprachiasmatic nucleus of the hypothalamus
TT	Taenia tecta
3 V	Third ventricle
VPL	Ventral posterolateral nucleus of the thalamus
VMH	Ventromedial hypothalamic nucleus
VPall	Ventral pallidum
VIS	Visual areas of the cortex
ZI	Zona incerta

Introduction

The nonapeptides oxytocin (OXT) and vasopressin (AVP) modulate numerous social behaviors ranging from affiliation and pair bonding to anxiety and aggression (Albers 2012; Donaldson and Young 2008; Veenema and Neumann 2008; Rigney et al. 2022). OXT and AVP are produced in distinct populations in the hypothalamus, primarily those of the paraventricular and supraoptic nuclei, as well as in extra-hypothalamic regions, and act on the OT-receptor (OXTR) and AVP1a-receptor (AVPR1A) via direct projections and

dendritic release (Kelly and Goodson 2014; Ludwig and Leng 2006; Rood and De Vries 2011). Although the nonapeptide system is strongly conserved across vertebrates, there is substantial variation in receptor distributions and densities. Comparative and experimental studies have revealed that differences in OXTR and AVPR1A neuroanatomical distributions reflect differences in social phenotype observed both across and within species (Caldwell 2017; Goodson et al. 2009; Young and Wang 2004; Campbell et al. 2009). For example, differences in nonapeptide receptor densities have been causally linked to differences in the expression of monogamy-related behaviors, such as pair bonding, within and across monogamous and non-monogamous *Microtus* vole species (Barrett et al. 2013; Lim et al. 2004; Walum and Young 2018). Additionally, while some species, such as the prairie voles (Froemke and Young 2021), exhibit dense OXTR binding in the nucleus accumbens (NAcc), a region critical for social reward (Liu and Wang 2003), laboratory mice and social tuco-tucos exhibit no detectable OXTR binding in this region (Beery et al. 2008; Borie et al. 2022b). Because of this variation across species, it is important to characterize nonapeptide receptor distributions in a species prior to examining nonapeptide-mediated behavior.

Nonapeptide receptor distributions have been characterized in a multitude of rodents including prairie voles, pine voles, montane voles, meadow voles, Taiwan voles, tuco-tucos, naked mole rats, ground squirrels, laboratory rats, singing mice, California mice, Syrian hamsters, and laboratory mice (Beery et al. 2008; Bester-Meredith et al. 1999; Campbell et al. 2009; Chappell et al. 2016; Freeman et al. 2019; Insel et al. 1994; Kalamatianos et al. 2010; Mitre et al. 2016; Smith et al. 2017; Taylor et al. 2020; Young 1999; Inoue et al. 2022). However, OXTR and AVPR1A distributions have yet to be characterized in the spiny mouse (*Acomys cahirinus*), an emerging rodent model in the field of social neuroscience (Fricker et al. 2021; Gonzalez Abreu et al. 2022; Kelly and Seifert 2021).

Spiny mice were historically used as a disease model to study diabetes progression (Gonet et al. 1966; Shafrir 2000), but have emerged as a model for studying complex tissue regeneration given their remarkable ability to regenerate skin and musculoskeletal (Gawriluk et al. 2016; Seifert et al. 2012). Beyond organ regeneration, these rodents also exhibit unique cytoprotective features that provide enhanced repair of major internal organs and the central nervous system (Koopmans et al. 2021; Nogueira-Rodrigues et al. 2022; Okamura et al. 2021; Peng et al. 2021; Saxena et al. 2019). Spiny mice have also become an excellent model for studying human-like reproductive biology, particularly their demonstration of a menstrual cycle (Bellofiore et al. 2017, 2018). Furthermore, spiny mice are also an exciting model for developmental studies given their precocial mode of development (Brunjes 1990; Haughton et al. 2016).

Spiny mice are cooperative breeders that live in large colonies nested within rocky outcroppings in Africa, the Middle East, and southern Asia (Deacon 2009; Frynta et al. 2011; Nowak 1999). They are socially bold, highly prosocial, and exhibit little aggression in both reproductive and nonreproductive contexts with both novel and familiar conspecifics (Fricker et al. 2021; Gonzalez Abreu et al. 2022). Spiny mouse males and females provide alloparental care to unrelated neonates at an equal rate to that of which they provide their own offspring (Porter et al. 1980; Tuckova et al. 2016), and established breeding groups will welcome a newcomer into the group; a behavior that is relatively rare among rodents (Cizkova et al. 2011). Spiny mice also exhibit a strong preference to affiliate with larger over smaller groups (referred to as gregariousness) (Fricker et al. 2021; Gonzalez Abreu et al. 2022), and thus are an excellent model for studying associative grouping and the relevant neural mechanisms that underlie these behaviors.

We previously mapped the distribution of nonapeptide-producing neuronal populations in spiny mice (Kelly and Seifert 2021); however, there remains a need to characterize distributions of nonapeptide receptors in this species. Thus, here we performed receptor-binding autoradiography and duplex in situ hybridization for OXTR and AVPR1A binding and *Oxtr* and *Avpr1a* mRNA to map the distribution of nonapeptide receptor binding and mRNA throughout the basal forebrain and midbrain of male and female spiny mice. Identifying sites of binding provides insight into where nonapeptides act upon functional receptors, whereas identifying sites of mRNA expression tells us where those receptors are synthesized.

In contrast, if binding is detected in regions where mRNA is not, it suggests the receptors detected in that area originate in another area and are transported down axons and act presynaptically (Inoue et al. 2022). Since nonapeptide receptor expression can be sexually dimorphic (Sharma et al. 2019; Young 1999), we also sought to examine associations with gonadal sex on OXTR and AVPR1A binding densities, focusing specifically on brain regions involved in social behavior and reward—collectively known as the Social Decision-Making Network (SDMN) (O'Connell and Hofmann 2011, 2012)—for which distributions of OXTR and AVPR1A receptors have been found throughout for many rodents (Albers 2015; Caldwell 2017; Smith et al. 2017).

Methods

Animals

All spiny mice in this study were derived from breeders obtained from the Seifert Lab at the University of Kentucky and maintained at Emory University. Sex was defined based

on gonads, with males having testes present and females having ovaries present. Animals were housed in groups of 2–5 same-sex siblings in either a GR1800 Double Decker cages (Tecniplast, West Chester, PA, USA) or a standard polycarbonate rat cage. They were provided ad libitum water and Purina Prolab RMH 1000 (Lab Diet, St. Louis, MO, USA). The colony room was maintained at an ambient temperature of $24\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$ and kept on a 14 h:10 h light/dark cycle (lights on at 7:00 AM). All procedures were approved by the Institutional Animal Care and Use Committee of Emory University (Protocol 201,900,126).

Tissue preparation

All subjects were adults (female $n=8$; male $n=8$) aged between PND 60 and 250, with age spans equally representing the entire range for both males and females. Subjects were euthanized by CO_2 inhalation, and their brains were dissected and flash-frozen on powdered dry ice. The brains were then wrapped in tinfoil and stored in a $-80\text{ }^{\circ}\text{C}$ freezer. Brains were thawed to $-20\text{ }^{\circ}\text{C}$ and coronally cryosectioned at $20\text{ }\mu\text{m}$ thickness. Multiple series of each brain were collected at $200\text{ }\mu\text{m}$ intervals on to Superfrost Plus slides (Fisher Scientific Co., Pittsburgh, PA, USA). Brain-mounted slides were again stored in the $-80\text{ }^{\circ}\text{C}$ freezer until further processing.

Receptor autoradiography receptor-binding assays

Receptor binding autoradiography for OXTR and AVPR1A were performed as previously described (Ross et al. 2009) on two sets of tissue from each brain. Briefly, slides were thawed and then post-fixed in 0.1% paraformaldehyde for 2 min. They were washed in two changes of 50 mM Tris buffer for 10 min each and transferred to binding chambers where they were immersed in I-125 Tracer Buffer for 1 h. Slides labeled for OXTR and AVPR1A were exposed to 50 pM ^{125}I -ornithine vasotocin analog (I-125 OVTA, PerkinElmer, Waltham, MA, USA) and 50 pM ^{125}I -Linear V1a antagonist (I-125 LVA, Perkin Elmer, Waltham, MA, USA), respectively. Slides were then washed at $4\text{ }^{\circ}\text{C}$ in four changes of Tris and Magnesium Chloride Buffer for 5 min each. They were then dipped in sterile ultra-pure deionized water and dried completely. Slides labeled for AVPR1A binding were exposed to BioMax MR film (Kodak, Rochester, NY, USA) for 4 days. Slides labeled for OXTR binding were exposed to film for 10 days.

Receptor autoradiography competitive binding assay

Competitive binding assays were performed using six series of adjacent tissue, three for the OXTR competitive binding

assay and three for the AVPR1A competitive binding assay. For both assays, 50 nM of unlabeled OXTR-selective agonist [Thr4, Gly7]OT (TGOT) (Bachem, Torrance, CA, USA) or Manning compound (Bachem, Torrance, CA, USA) was applied. Specific binding of I-125 OVTA was eliminated by the application of TGOT, but was not affected by the application of Manning Compound, confirming that I-125 OVTA is specific to the OXTR in the spiny mouse. Specific binding of I-125 LVA was only partially blocked by the application

of Manning Compound and was not affected by the application of TGOT (Fig. 1).

Receptor autoradiography imaging and quantification

Films were placed on the Northern Light B95 Imaging Research Precision Illuminator (Imaging Research Inc., St. Catharines, Ontario, Canada) and imaged using a 55 mm

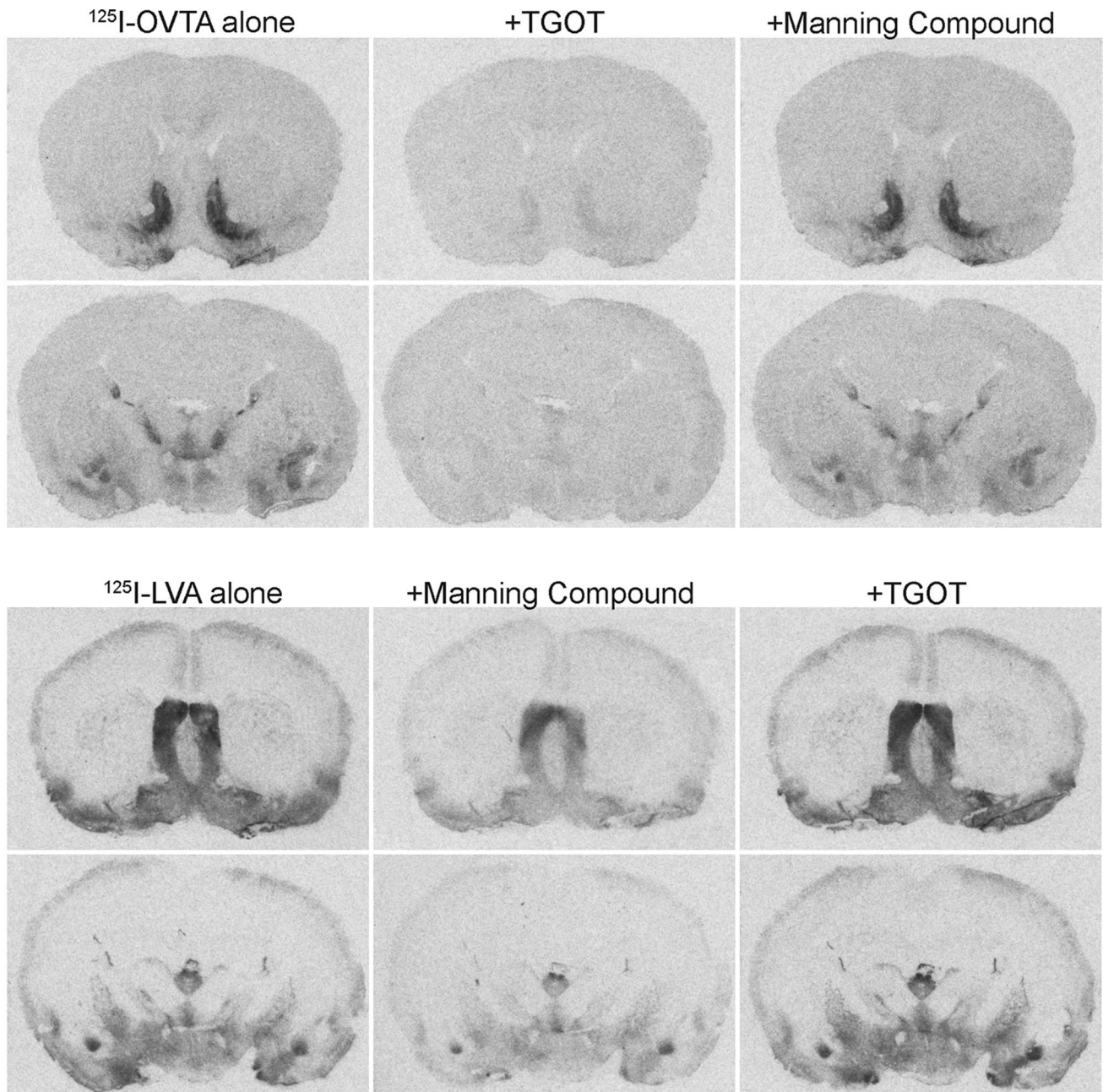


Fig. 1 Autoradiograms of receptor binding in the presence or absence of receptor antagonists. Application of TGOT, an oxytocin receptor antagonist, largely blocks I-OVTA binding in the spiny

mouse. (Bottom) Application of Manning Compound, a V1a receptor antagonist, only partially blocks I-LVA binding in the spiny mouse

Nikon lens connected to QCapture Pro imaging software (Photometrics, Tucson, AZ, USA). All images were captured in the same sitting using the same illumination, focus, and exposure settings.

We sought to identify sex differences in brain regions involved in social behavior and reward and therefore focused on nodes within the SDM, which included: nucleus accumbens (NAcc), lateral septum (LS), ventral pallidum (VPall), medial preoptic area (MPOA), bed nucleus of the stria terminalis (BNST), anterior hypothalamus (AH), basolateral amygdala (BLA), medial amygdala (MeA), caudate putamen (CPu), ventromedial hypothalamus (VMH), hippocampus (Hi), and periaqueductal gray (PAG). Regions of interest (ROIs) were identified by comparing dark regions on the autoradiographs to the Paxinos and Franklin Mouse Atlas, 5th edition (Paxinos and Franklin 2012) and images of the same or adjacent sections counterstained for Nissl. Receptor binding density was quantified in a semi-automated fashion using a custom FIJI macro (Schindelin et al. 2012). In this pipeline, images were automatically converted to an 8-bit grayscale format, with pixel grayscale values ranging from 0 to 255, where a value of 0 represents a black pixel (high radioactivity) and a value of 255 represents a white pixel (low radioactivity). ROIs were manually outlined using FIJI's freehand tool and then automatically renamed, measured, and saved based on user-provided information about the subject ID and ROI. Up to four measurements per subject were taken, two from the left and two from the right hemisphere, from intact sections capturing the ROI.

One ROI was used to determine background levels for the entire brain of each individual subject within an assay. To identify an appropriate background ROI, each image was pseudo-colored using the 16 color LUT. This allowed us to identify areas of intact gray matter that had the highest grayscale values (i.e., the least amount of radioactivity). Once identified, receptor binding density within the ROI was measured as described above. Visual cortex and the hippocampus were used as background ROIs for the OXTR and AVPR1A assays, respectively.

Each binding measurement was adjusted for background by taking the absolute value of the difference between the ROI's grayscale value and the subject-specific background value, resulting in a positive value. Then, an average for each ROI within each subject was taken. For the resulting data, a higher final grayscale value indicated higher receptor binding density.

In situ hybridization

Duplex in situ hybridization was performed on one set of tissue from the brains of three females and one male used in the autoradiography assays following a modified version of the RNAscope Multiplex Fluorescent Reagent Kit v2 Assay

for Fresh-frozen Samples (technical bulletin #323,100-USM, Advanced Cell Diagnostics, Newark, CA, USA). Briefly, sections were fluorescently double-labeled for *Oxtr* and *Avpr1a* mRNA using RNAscope Probe—Ad-Oxtr (Part ID: 819,661-C2) and RNAscope Probe—Ad-Avpr1a (Part ID: 821,881), respectively. Probes were designed by ACD Inc. based off the following gene sequences from the spiny mouse genome:

Arginine vasopressin 1a receptor gene

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ATGAGTTTCCCGCGAGGCTCCACGATCGGCCC
GCGAGCAACTCCAGCCCGTGGTGGCCTCTGATC
GCCGCCGAGGGTGTCAACGCCAGCCGGGAGGCA
TCCGGGCTTGGGGAAGGTAGCGGCCCGCCAGGG
GATGTGCGCGATGAGGAGCTGGCCAAGGGGAA
ATCGCTGTGCTGGCGGTGATTTTCGTGGTGGCTGTG
CTGGGCAATAGCAGTGTGCTGCTGGCGCTGCATCGC
ACGCCTCGCAAGACGTCCCGCATGCATCTCTTCATC
CGACACCTCAGCCTGGCAGACCTGGCGGTAGCCTTC
TTCCAAGTGCTGCCACAGCTGTGCTGGGACATCACC
TACCGCTTCCCGGGCCGCGACTGGCTGTGCCGCGTG
GTGAAGCACCTGCAGGTGTTTGCTATGTTTCGCGTCT
GCCTACATGCTGGTGGTCATGACCGCGGACCGCTAC
ATCGCGGTGTGCCACCCGCTCAAGACCCTGCAG
CAGCCTGCGCGCCGCTCGCGCCTCATGATCGCTGCC
TCCTGGGTGTTGAGTTTCTACTGAGCACGCCACAG
TACTTCATCTTCTCTATGATCGAAATCGAGGTGAAC
AACGGTACCAAACTCAAGACTGCTGGGCTACCTTC
ATCCAGCCTTGGGGTCCCCGCGCCTACGTGACCTGG
ATGACCAGCGGTGTCTTTGTGGCACCTGTGATCGTC
TTGGGTACCTGCTATGGCTTCATCTGTTACCACATC
TGGAGCAATGTCCGCGGGAAGACGGTGTGCGGA
CAGAGCAAGGGCAGCAAGGGCTCTGGGGAAGCC
TTGGGTCCCTTCCACAAGGGGCTTCTGGTCACGCCT
TGTGTCAGCAGCGTGAAGACCATTTCGCCGCCAAG
ATCCGCACAGTGAAGATGACCTTTGTCATCGTAACT
GCATACATCCTCTGCTGGACGCCTTTCTTCATCGTC
CAGATGTGGTCCGTCTGGGATGACGATTTCGCCTGG
ACCGATTAGAAAATCCTTCCATAACCATCACCGCG
TTACTGGCCTCCTTGAACAGCTGCTGCAACCCGTGG
ATATACATGTTTTTTAGTGGCCACCTCCTGCAAGAC
TGTGTCCAAAGTTTGCCGTGCTGCCAAGCGTGGCT
CAGAAATTCACCAAGGACGACTCAGACAGCATG
AGCCGAAGACAGACTTCGTACTCCAACAACCGA
AGCCCAACAAACAGCATCGGGACATGGAAGGAC
TCGCCTAAATCTTCCAAGTCCATCAGGTTTCATTCT
GCTTCCACCTGA.
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Oxytocin receptor gene

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ATGGAGGGCACGCCGGCAGCCAACTGGAGCATC
GAGTTGGACTTCGGGAGTGGAGTGCCACCAGGG
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GTGGAGGGAAACCGCACAGCCGGTCCGCCAAGG
CGCAACGAGGGCCCTGGCGCGGGTGGAGGTGGCG
GTGCTGTGCCTCATTCTGTTCTGGCACTGAGCGGC
AACGCGTGCGTGCTGCTGGCTCTGCGCACACAG
CGCCACAAACATTTCGCGTCTTCTTTTCATGAAG
CACCTGAGCATCGCCGACTTGGTGGTGGCTGTGTTC
CAGGTGCTCCCGCAGCTGCTGTGGGACATCACCTTT
CGCTTCTACGGGCCCCGACTTGCTGTGTCGTCTGGTC
AAATACTTGCAGGTGGTGGGCATGTTGCGCTCTACT
TACCTGTTGCTGCTCATGTGCTCGACCGCTGCCTG
GCCATTTGCCAGCCTCTGCGTTCGCTGCGCCGCCGA
ACGGACCGCCTGGCGGTGCTGGCGACGTGGCTC
GGCTGCCTAGTGGCTAGCGCGCCGCAGGTGCAC
ATTTTCTCACTGCGCGAAGTGGCAGACGGCGTCTTT
GATTGCTGGGCGGTCTTCATCCAGCCCTGGGGACCC
AAGGCCTACGTACGTGGATCACGCTTGCTGTCTAC
ATTGTGCCTGTGCATCGTGGCCGCTGCTATGGC
CTCATCAGCTTCAAGATCTGGCAGAACCTGCGGCTC
AAGACGGCAGCAGCGGCTGCTGCAGCCGAGGGG
AGTGATGCCGGCGGAGGTGGGCGTGCGGCACTG
GCAAGGGTCAGTAGTGTCAAGCTCATCTCCAAG
GCCAAGATCCGCACAGTGAAGATGACCTTCATTATC
GTACTGGCCTTCATCGTCTGCTGGACGCCTTTCTTC
TTCGTGCAGATGTGGAGCGTCTGGGATGTCAATGCG
CCCAAGGAAGCGTCTGCCTTCATCATTGCCATGCTC
TTGGCCAGCCTCAACAGCTGCTGCAACCCCTGGATC
TACATGCTCTTCACAGGTCACCTCTTCCACGAACCT
GTGCAGCGCTTCTCTGCTGCTCCGCCCGGTACCTG
AAGGGCAGTCGGCCCCGGAGAAACAAGCATCAGC
AAGAAAAGCAACTCATCCACCTTTGTCCTGAGTCGT
CGCAGCTCCAGCCAGCGGAGCTGTTCTCAGCCATCC
TCAGTGTGA.

The full protocol was as follows. Tissue was fixed for 60 min at room temperature in 4% paraformaldehyde in 0.1 M phosphate buffer saline (PBS) and then rinsed twice in 0.1 M PBS. The tissue was then dehydrated in 50%, 70%, and two changes of 100% ethanol for 5 min each. Slides were air dried for 5 min before applying a hydrophobic barrier using an ImmEdge Hydrophobic Barrier Pen (Vector Laboratories, Burlingame, CA, USA). Each section was completely covered in hydrogen peroxide (600–1800 μ L per slide) and incubated in the HybEZ Humidity Control Tray at room temperature for 10 min. Slides were washed twice in deionized water. Then, 300 μ L of Protease III was pipetted on to each slide. Slides were incubated at room temperature in the HybEZ Humidity Control Tray for 5 min and then rinsed twice in 0.1 M PBS. Next, 300 μ L of the appropriate probe mix was added to each slide. The experimental probe mix consisting of RNAscope Probe—Ad-Oxtr and RNAscope Probe—Ad-Avpr1a was added to the experimental slides. The probe mix for the positive control slide was RNAscope Probe—Mun-Polr2a and the probe mix for the negative control slide was RNAscope Probe—Mun-Ppib-C2.

In situ hybridization imaging and mRNA expression analysis

Slides were imaged using the Vectra Polaris slide scanner (Akoya) within one week of staining. An autofluorescence channel was captured during scanning. We measured the percentage of cells within a ROI that expressed mRNA using the Fluorescent RNAscope Image Analysis in QuPath protocol (technical note MK 51–154, Advanced Cell Diagnostics, Newark, CA, USA).

Briefly, slide scanner images were imported as fluorescent images into a project in QuPath (Bankhead et al. 2017). Each section was then “imaged” from these scanner images by drawing a rectangular annotation around the entirety of each section, sending the region to ImageJ, and saving the annotation as its own file. In doing so, each image was then small enough to be processed in FIJI. In FIJI, the background subtraction tool was used on each channel (DAPI, *Oxtr*, *Avpr1a*, and autofluorescence) with a rolling ball radius of 50. Then, the autofluorescence channel was subtracted from the *Oxtr* and *Avpr1a* channels to remove noise. This removed any objects that co-occurred in both the autofluorescence and signal channels and corrected the uneven autofluorescence between and across subjects.

ROIs identified during the autoradiography assay were assessed for mRNA expression. In QuPath, two annotations of equal size, one on the left hemisphere and one on the right, were placed on the ROI using the Specify Annotation function. Then, the Cell Detection module was used to segment cells within the annotation box stained with DAPI. The following parameter values were used: requested pixel size of 0.5 μ m, background radius of 10 μ m, median filter radius of 0 μ m, sigma of 1.5 μ m, minimum area of 10 μ m², maximum area of 400 μ m², and cell expansion of 5 μ m. ‘Split by shape’, ‘include cell nucleus’, ‘smooth boundaries’, and ‘make measurements’ were all selected. To account for uneven DAPI staining, a unique threshold was determined for each image by measuring from ten background regions, consistent with the DEFiNE quantification method (Powell et al. 2019). The average mean and standard deviation across the ten measurements were calculated and the threshold was set at 4 standard deviations above the mean. In our dataset, this resulted in threshold values ranging from 3 to 28.

Once the cells were segmented, the Subcellular Detection Module was used to detect the fluorescently labeled probes within cell boundaries. The following parameter values were used: expected spot size of 2 μ m², min spot size of 0.5 μ m², and a maximum spot size of 6 μ m², following the Technical Note’s recommended settings. ‘Include clusters’ was selected. Like with the DAPI channel, a unique threshold was determined for each channel based on the average background. For *Oxtr* mRNA, a threshold ranging from 1 to

3 was applied and for *Avpr1a* mRNA, the thresholds ranged from 1 to 4.

Detection measurements were exported from QuPath into Excel for summarization. To minimize noise, a cell needed to have at least two puncta detected within its boundaries. Data were quantified to reflect the percentage of cells that express mRNA within a ROI. The sample size was too small to perform any statistical analyses and the subsequent results will be described qualitatively by binning each individual's results into the following ranges: 1–25% (+), 26–50% (++), 51–75% (+++), 76–100% (++++) (as described in Inoue et al. 2022).

Nissl staining procedure

The slides used for OXTR autoradiography were counter stained using toluidine blue Nissl. Briefly, slides were submerged in the dark in 180 mL of deionized water with 500 μ L of 0.125 M toluidine blue stock solution. Then, the tissue was rinsed in deionized water and dehydrated in 70% ethanol for 20 s, 95% ethanol for 30 s, and two rinses of 100% ethanol for 3 min each. Then the slides were submerged in two rinses of Xylene for 5 min each. Slides were cover slipped using DPX. Slides were imaged using an Epson Perfection V700 photo scanner (Epson, Suwa, Nagano, Japan) in RGB mode at 2500 dpi.

Statistics

All statistical analyses were performed in SPSS 28 (IBM Analytics, USA). We were interested in characterizing sex differences in functional protein within regions of the SDMN and therefore analyzed tissue processed via receptor autoradiography. Data were normally distributed, and thus we compared receptor binding between sexes in brain

regions in the SDMN when binding was present using independent samples *t* tests. The false discovery rate (FDR) procedure was used to correct for multiple comparisons, which were calculated separately for OXTR and AVPR1A. Corrected *p* values are reported in the figures. Both uncorrected and corrected *p* values are in Tables 1 and 2. Figures were made using Prism 9 (GraphPad, USA). We did not examine sex differences in mRNA expression due to limitations of RNAscope. The processing system (i.e., HybEZ™) can accommodate a maximum of 20 slides. With the need for control slides, this limits the number of experimental slides that can be processed in a single run. To process tissue for nearly whole brains and have a sample size sufficient to detect associations with gonadal sex, multiple runs would be needed. Because there can be substantial variation in mRNA labeling across runs, it would be difficult to disentangle associations with gonadal sex from effects of batch/run. Additional analyses described as follows were conducted in R version 4.0.5 (2021-03-31) (Team 2013). To visualize

Table 2 Statistical details of associations with gonadal sex for AVPR1A binding densities in regions of the Social Decision-Making Network of spiny mice

Brain region	T(df)	<i>p</i> value uncorrected	<i>p</i> value FDR corrected	Effect size (Cohen's <i>d</i>)
LS	T(14)=1.02	0.33	0.66	1.30
VPall	T(13)=2.01	0.06	0.26	0.48
MPOA	T(14)=0.48	0.64	0.85	0.01
BNSTa	T(12)=1.21	0.26	0.68	0.25
AH	T(14)=2.47	0.03	0.23	0.14
MeA	T(14)=0.56	0.59	0.94	0.47
VMH	T(14)=0.01	0.99	0.99	0.85
PAG	T(14)=0.18	0.86	0.98	0.10

Table 1 Statistical details of associations with gonadal sex for OXTR binding densities in regions of the Social Decision-Making Network of spiny mice

Brain region	T(df)	<i>p</i> value uncorrected	<i>p</i> value FDR corrected	Effect size (Cohen's <i>d</i>)	Direction of sex effect
NAcc	T(14)=1.16	0.27	0.30	0.58	
LS	T(14)=3.21	<0.01	0.02	1.62	Higher in males
VPall	T(14)=1.32	0.210	0.26	0.66	
MPOA	T(14)=1.15	0.27	0.29	0.58	
BNSTa	T(14)=5.78	<0.01	<0.01	2.91	Higher in males
BNSTp	T(14)=7.29	<0.01	<0.01	3.65	Higher in males
AH	T(14)=5.76	<0.01	0.01	2.88	Higher in males
BLA	T(14)=0.89	0.39	0.39	0.45	
MeA	T(14)=1.96	0.07	0.11	0.98	
CPu	T(14)=2.32	0.04	0.08	1.16	
VMH	T(14)=1.68	0.12	0.17	0.84	
Hi	T(14)=2.23	0.04	0.08	1.12	

Significant effects after correction for multiple comparisons are bolded

binding density patterns across brain regions, we used Pearson correlation matrices calculated via the R function “cor” to generate heatmaps via the function ‘heatmap.2’ in the R package ‘gplots.’ We then conducted hierarchical clustering analyses on the binding density data using the R package ‘pvcust’ (Suzuki et al. 2019) with distance metric set to ‘correlation’, linkage metric set to ‘average’, and bootstrap values set to 1000. Significant clusters were defined as those with an approximately unbiased (AU) p value > 95 .

Results

Mapping of OXTR and AVPR1A binding in the basal forebrain and midbrain

To map expression of OXTR and AVPR1A binding in the brains of male and female spiny mice, we conducted receptor autoradiography on brain sections extending from the olfactory bulbs to the periaqueductal gray. We observed I-125 OXTR and I-125 AVPR1A binding across several brain regions, namely olfactory areas, cortical areas, hippocampal areas, striatal areas, the pallidum, the amygdala, the hypothalamus, the thalamus, and the midbrain. Representative autoradiograms demonstrating receptor distributions are provided for OXTR in Fig. 2 and AVPR1A in Fig. 3.

Sex effects in brain regions involved in social behavior and reward

To determine if gonadal sex influences nonapeptide receptor densities as has been observed for other species (Dumais

et al. 2013), we examined associations with gonadal sex on receptor binding in brain regions in the SDMN. We observed OXTR binding throughout more regions of the SDMN compared to AVPR1A binding. Additionally, independent t -tests revealed an association of gonadal sex on OXTR binding in the LS, BNST, and AH, with males having higher binding densities than females (all $p < 0.02$; Fig. 4A; Table 1). We previously found nonapeptide-producing neuronal sex differences in anatomy (Kelly and Seifert 2021) and function (Kelly unpub. obs) on an anterior–posterior axis of the BNST population of spiny mice, and therefore analyzed anterior and posterior portions of the region separately here. However, we observed similar findings for both compartments of the BNST. We found no associations with gonadal sex on any AVPR1A binding sites within the SDMN (all $p > 0.23$; Fig. 4B; Table 2).

Co-expression patterns of OXTR and AVPR1A binding in the SDMN

Because brain regions within the SDMN are anatomically interconnected and may coordinate activity to modulate distinct social behaviors (Deng et al. 2019; Dong and Swanson 2006; Newman 1999; O'Connell and Hofmann 2011), we ran correlational analyses for OXTR and AVPR1A binding for each sex to explore relationships between binding densities in these regions. To assess correlated OXTR and AVPR1A binding expression between pairs of regions, we first conducted Pearson correlation tests. The resulting correlation coefficients were visualized using a heatmap; due to the exploratory nature of the analysis, we did not correct for multiple comparisons. While males showed a mix of positive

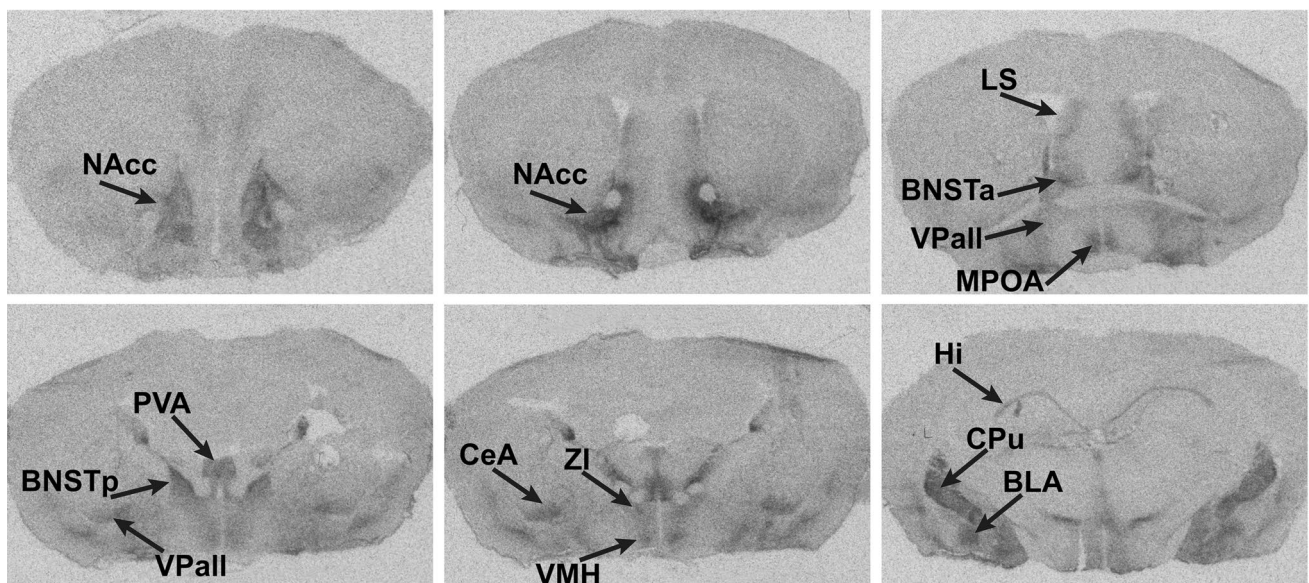


Fig. 2 OXTR binding. Representative autoradiograms of OXTR binding in coronal sections from spiny mouse tissue

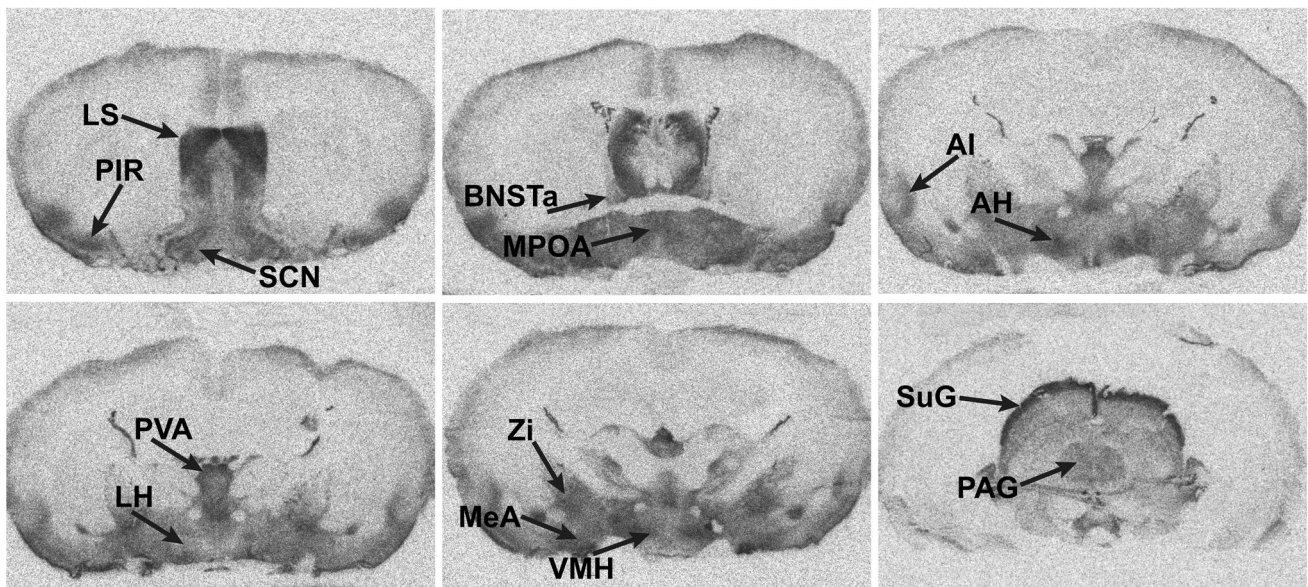
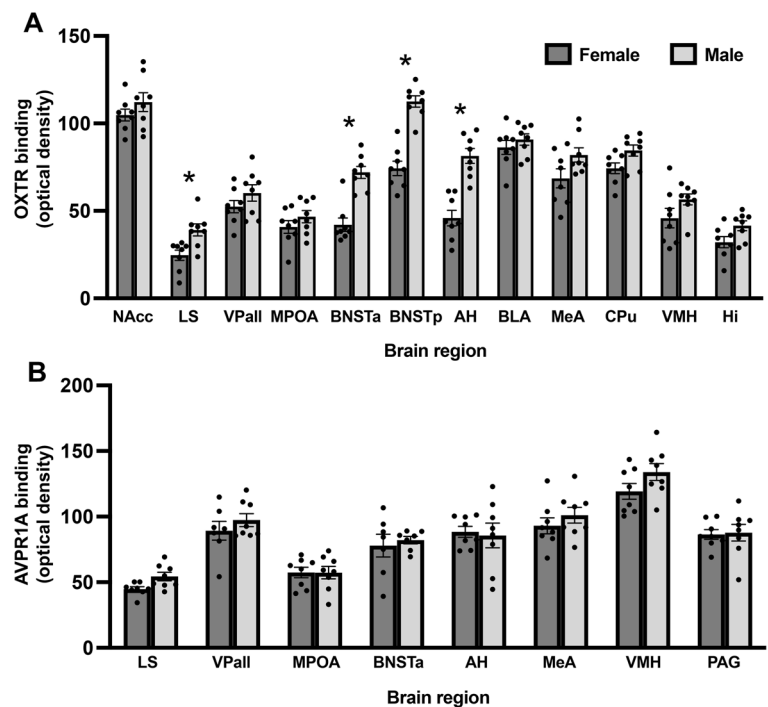


Fig. 3 AVPR1A binding. Representative autoradiograms of AVPR1A binding in coronal sections from spiny mouse tissue

Fig. 4 Associations with gonadal sex on OXTR and AVPR1A binding. Mean grayscale values ($\pm$ SEM) for **A** OXTR and **B** AVPR1A binding in brain regions of the Social Decision-Making Network (SDMN) in male and female spiny mice. *Denotes $p_{\text{corrected}} < 0.05$



and negative correlations in OXTR binding between brain regions across the SDMN, females showed primarily positive correlations (Fig. 5A). However, the opposite pattern emerged for AVPR1A binding (Fig. 5B).

To further explore patterns of co-expression across SDMN regions, we performed hierarchical clustering analyses using the R package ‘pvclust’ (Suzuki et al. 2019) with distance metric set to ‘correlation’, linkage metric set to ‘average’, and bootstrap values set to 1000.

Significant clusters were defined as those with an approximately unbiased (AU) p value > 95 . Again, the two sexes exhibited distinct co-expression patterns. For OXTR binding in male spiny mice, hierarchical clustering identified a large cluster comprised of the NAcc, MeA, BNSTp, VPall, BLA, LS, CPu, BNSTa, and Hi, whereas in females we observed a large cluster consisting of the BNSTa, BNSTp, VMH, VPall, BLA, and Hi, as well as a small cluster comprised of the MeA and AH (Fig. 5A). Despite observing

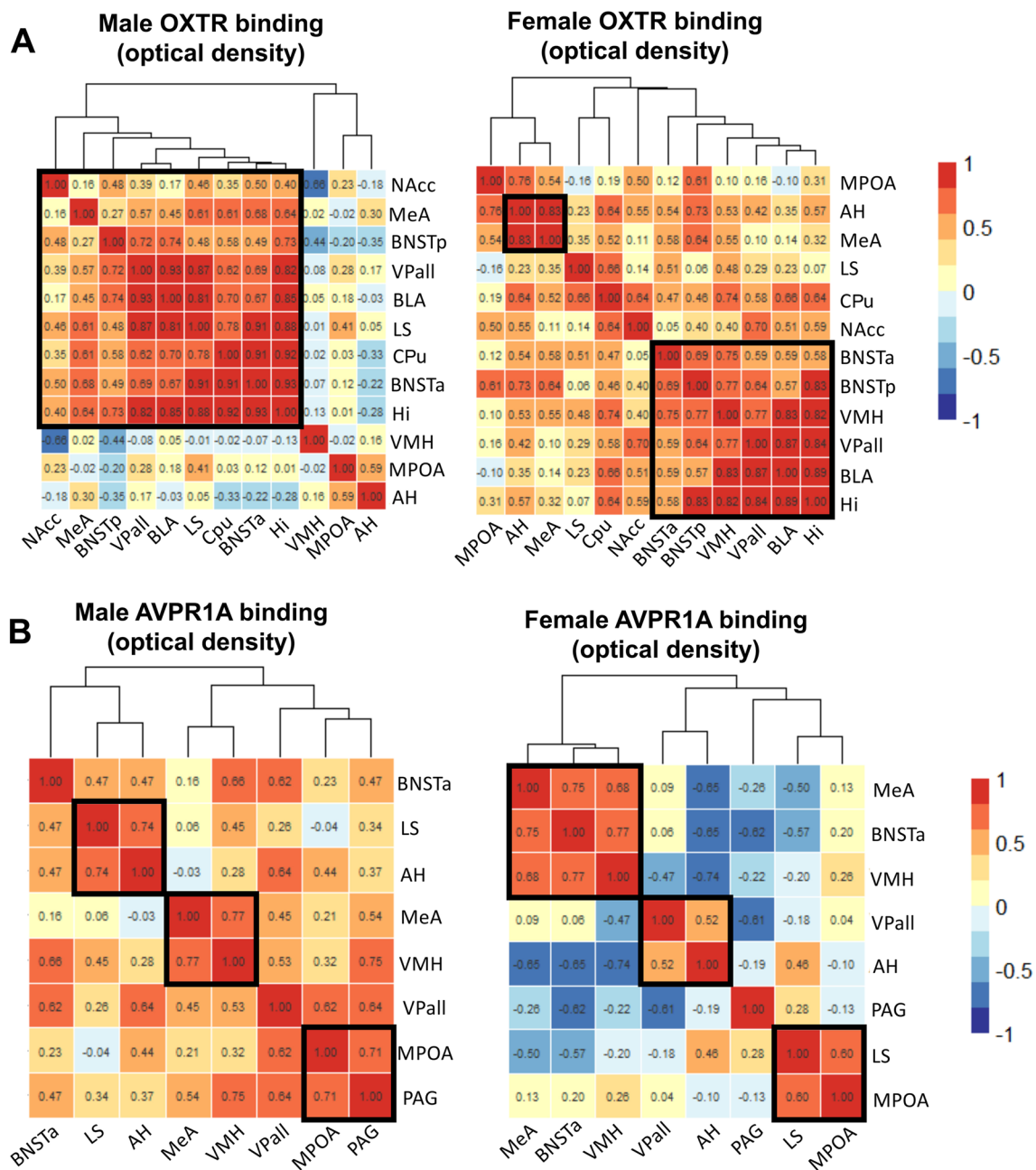


Fig. 5 Pairwise comparisons and hierarchical clustering of **A** OXTR and **B** AVPR1A binding expression across regions of the SDM. Values shown are Pearson correlation coefficients. Significant clus-

ter (AU > 95) are outlined in black boxes. Due to small sample sizes, note that this is exploratory in nature

no sex differences in AVPR1A binding in individual brain regions, the co-expression patterns for AVPR1A binding across the SDM are strikingly different for males and females. Hierarchical clustering revealed 3 clusters for males: (1) LS and AH, (2) MeA and VMH, and (3) MPOA and PAG, and 3 clusters for females: (1) MeA, BNSTa, and VMH, (2) V Pall and AH, and (3) LS and MPOA (Fig. 5B).

Mapping of *Oxtr* and *Avpr1a* mRNA in the basal forebrain and midbrain

To investigate expression of *Oxtr* and *Avpr1a* mRNA in the spiny mouse brain, we performed RNAscope in situ hybridization in 2 male and 2 female brains on sections extending from the olfactory bulb through the periaqueductal gray. We

Table 3 *Oxtr* and *Avpr1a* mRNA expression across the forebrain and midbrain of the spiny mouse

Region	<i>Oxtr</i> mRNA			<i>Avpr1a</i> mRNA			<i>n</i>
	Bins	Mean (%)	SD	Bins	Mean (%)	SD	
<i>Olfactory areas</i>							
AOB	+/+ +	10.99	10.99	+ + + /+ + + +	79.28	9.56	4
AOA	+/+ + +	51.40	22.76	+/+ + +	26.38	24.69	4
MOBgl	+/+ +	20.06	8.31	+/+ + /+ + +	35.25	17.85	3
MOBgr	+/+ +	25.11	2.72	+/+ +	28.14	19.07	3
PIR	+	13.70	9.36	+	12.49	7.72	4
TT	+/+ + /+ + +	38.68	20.71	+/+ + +	39.38	22.60	4
<i>Cortex</i>							
AI	+/+ +	18.47	21.33	+/+ + /+ + +	28.76	19.59	4
Cg2	+/+ + +	22.88	21.29	+/+ +	28.69	13.20	4
Cg5	+/+ + /+ + +	36.97	19.04	+/+ +	21.63	12.54	4
DI	+/+ + +	28.08	22.64	+/+ + +	21.11	21.01	4
<i>Striatum</i>							
CPu	+/+ +	31.04	7.15	+	7.89	5.76	4
LS	+	8.22	5.01	+ + +	59.35	6.85	4
NAcc	+ +	38.41	6.17	+/+ +	17.65	16.55	4
OT	+/+ +	18.68	19.40	+/+ +	21.92	16.21	4
<i>Pallidum</i>							
BNSTa	+/+ + /+ + +	31.52	19.39	+/+ +	13.63	9.41	4
BNSTp	+/+ + /+ + +	39.56	22.22	+/+ +	13.17	13.52	4
VDB	+	7.02	1.79	+	13.24	3.16	4
GP	+/+ +	21.11	14.62	+/+ +	23.61	18.97	4
VP	+/+ +	17.94	10.71	+	15.58	9.08	4
<i>Hypothalamus</i>							
AH	+/+ +	20.56	12.35	+/+ +	27.84	10.90	4
LH	+	16.25	5.24	+/+ +	16.88	11.70	4
MM	+	11.91	7.74	+	13.85	4.49	4
MPOA	+/+ +	9.71	13.19	+/+ +	21.64	13.53	4
VMH	+/+ +	19.01	11.02	+	16.53	7.04	4
ZI	+/+ +	21.31	17.79	+	9.50	5.29	4
<i>Thalamus</i>							
LD	+/+ +	11.02	10.43	+	11.96	8.03	4
MG	+/+ +	9.95	13.42	+/+ +	17.85	13.31	4
PVA	+	14.68	8.40	+/+ + +	51.16	29.04	4
PVG	+	15.70	8.22	+/+ +	25.38	17.55	4
Re	+	10.59	6.72	+/+ +	36.01	17.31	4
VPL	+/+ + +	18.40	23.42	+/+ +	13.41	12.40	4
<i>Amygdala</i>							
AHAL	+/+ + + /+ + + +	45.49	38.16	+	11.86	10.48	4
BLA	+/+ +	32.58	10.25	+/+ +	15.84	10.39	4
CeA	+ + /+ + +	40.49	10.80	+	11.89	6.63	4
MeA	+/+ +	19.18	11.36	+	12.41	5.12	4
<i>Hippocampus</i>							
CA1py	+/+ +	7.68	12.13	+	4.75	4.62	4
<i>Midbrain</i>							
APT	+/+ +	13.97	20.29	+/+ +	22.96	20.05	4
IP	+	2.84	1.65	+/+ + /+ + +	39.88	15.20	4
PAG	+	6.65	3.79	+/+ +	11.55	11.95	4
PVG	+	15.70	8.22	+/+ +	25.38	17.55	4

Table 3 (continued)

Region	<i>Oxtr</i> mRNA			<i>Avpr1a</i> mRNA			<i>n</i>
	Bins	Mean (%)	SD	Bins	Mean (%)	SD	
SuG	+	6.11	6.54	+	15.34	12.35	4
SCm	+	7.41	9.68	+ / + +	8.80	6.78	4

Oxtr and *avpr1a* were labeled in the brain using duplex in situ hybridization, and the percentage of *Oxtr*+ and *Avpr1a*+ cells were calculated for each ROI. The data were binned into the following categories based on the percentage of cells expressing mRNA: + : $\leq 25\%$; + + : $25\% < x \leq 50\%$; + + + : $50\% < x \leq 75\%$; + + + + : $> 75\%$

Multiple bins reflect variability across subjects. Mean and standard deviations represent the percentage of cells that were positive for the relevant transcript

observed both *Oxtr* and *Avpr1a* mRNA in 42 brain regions (see Table 3 for qualitative results). Nonapeptide receptor mRNA expression was observed in all regions of both sexes. Particularly robust *Oxtr* labeling was observed in the NAcc, BNST, and AH (Fig. 6) and robust *Avpr1a* labeling was observed in the LS, MPOA, and Hi (Fig. 7). Unlike OXTR and AVPR1A binding, all brain regions within the SDMN expressed some *Oxtr* and *Avpr1a* mRNA. For example, we observed *Oxtr* mRNA, but no OXTR binding, in the PAG. Further, we observed *Avpr1a* mRNA, but no AVPR1A binding, in the NAcc, CPu, BLA, and Hi. This suggests that nonapeptide receptors may be synthesized in these regions but transported to other areas of the brain.

Discussion

Here we provide the first characterization of distributions of OXTR and AVPR1A binding and *Oxtr* and *Avpr1a* mRNA in the basal forebrain and midbrain of the adult spiny mouse. We examined the effect of sex on functional protein throughout brain regions of the SDMN, and found that OXTR binding densities in the LS, BNSTa, BNSTp, and AH were greater in males compared to females. However, we observed no associations with gonadal sex in AVPR1A binding in individual brain regions. Interestingly, exploratory correlational analyses and hierarchical clustering revealed distinct patterns of AVPR1A and OXTR co-expression for males and females, suggesting that nonapeptide receptors in neuroanatomically connected regions may differentially coordinate activity in male and female spiny mice.

Similarities and dissimilarities in SDMN OXTR and AVPR1A binding in spiny mice and other rodents

There is an extraordinary amount of variation in nonapeptide receptor distributions and densities across species (Beery et al. 2008; Olazabal and Sandberg 2020; Froemke and Young 2021). Although we characterized nonapeptide receptor binding and mRNA distributions throughout the basal

forebrain and midbrain of spiny mice, we focused analyses on regions within the SDMN. The SDMN comprises brain regions involved in multiple types of social behavior as well as those in the mesolimbic reward system (O'Connell and Hofmann 2011, 2012). The hypothesis of SDMN function is that the network evaluates the salience of stimuli to facilitate a context-appropriate behavioral response, thus allowing an animal to cope with challenges and take advantage of opportunities in an ever-changing social environment. The presence of nonapeptide receptors in SDMN regions has been reported for several rodents, with more extensive characterizations in prairie voles and Wistar rats (Prounis et al. 2018; Smith et al. 2017).

We observed OXTR and AVPR1A binding throughout most regions of the SDMN in male and female spiny mice. Notably, we did not detect OXTR binding in the PAG or AVPR1A binding in the BLA, CPu, Hi, and NAcc. Few studies have assessed OXTRs in the PAG, however, the lack of OXTR binding in the PAG of spiny mice is consistent with what has been reported for Wistar rats (Smith et al. 2017). Similarly, an absence of AVPR1A binding in the BLA and CPu has also been reported for prairie voles and Wistar rats (Prounis et al. 2018; Smith et al. 2017). While AVPR1A binding is largely absent in the Hi of prairie voles and Wistar rats (Prounis et al. 2018; Smith et al. 2017), the dentate gyrus (DG) of the Hi typically shows robust OXTR binding in rats and tuco-tucos (Beery et al. 2008; Smith et al. 2017); yet, we observed no AVPR1A binding in the Hi or DG of spiny mice. Although, like spiny mice, AVPR1A binding in the NAcc is largely absent in prairie voles (Prounis et al. 2018), AVPR1A binding is present in a small portion of the shell of the NAcc in Wistar rats (Smith et al. 2017). Conversely, we observed moderate OXTR binding in the NAcc of spiny mice. The OXTR population of the NAcc is arguably one of the most well-studied nonapeptide receptor sites and exhibits a great deal of variation in binding densities even within a species. NAcc OXTR distinguishes mating system in monogamous and non-monogamous voles, relates to alloparental care and pairbond formation in prairie voles, regulates various aspects of social bonding and alloparental behavior in voles, and facilitates social reward

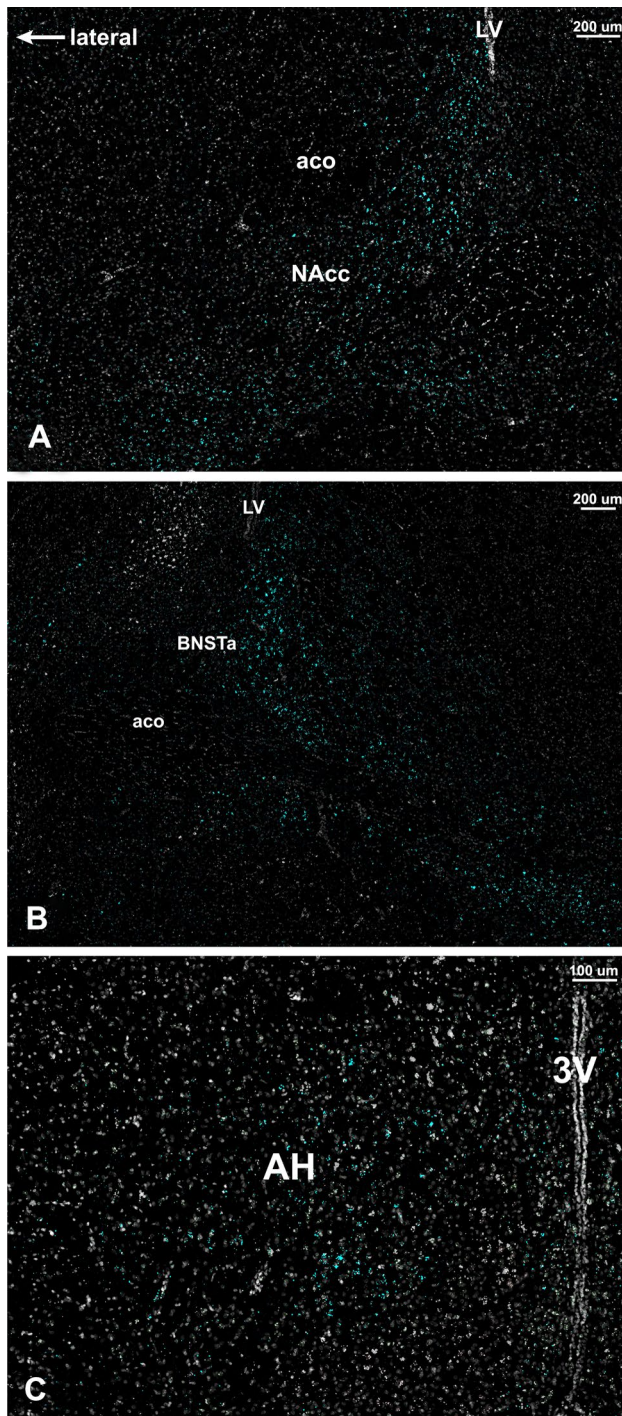


Fig. 6 *Oxt* mRNA expression in the spiny mouse. Representative images of *Oxt* mRNA expression from a female spiny mouse brain in the **A** NAcc **B** BNSTa, and **C** AH

processing in C57/BL6 mice (Dolen et al. 2013; Liu and Wang 2003; Borie et al. 2022a; Bosch et al. 2016; Keebaugh et al. 2015; Barrett et al. 2015). Interestingly, the spiny mice showed a unique expression pattern compared to other species such that OXTR binding was largely restricted to the

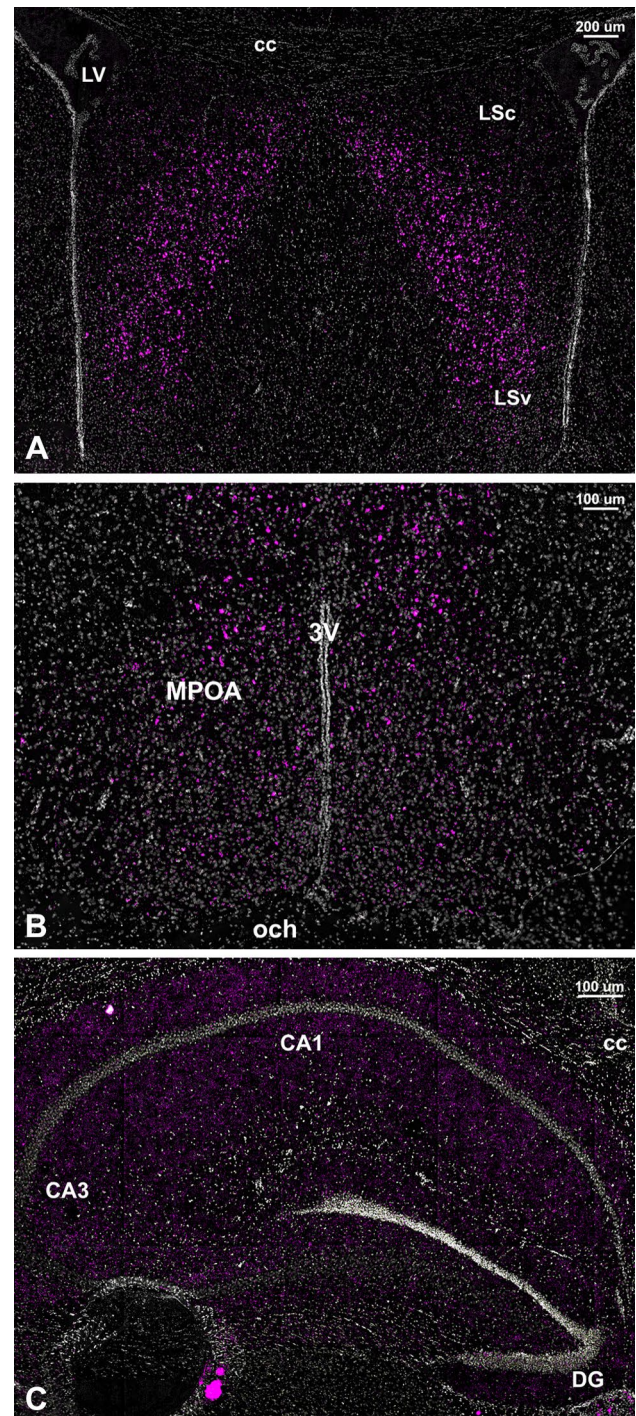


Fig. 7 *Avpr1a* mRNA expression in the spiny mouse. Representative images of *Avpr1a* mRNA expression from a female spiny mouse brain in the **A** LS, **B** MPOA, and **C** Hi

shell. Previous research has shown that the NAcc core and shell can differentially influence behaviors such as reward- and drug-seeking and learning (Chaudhri et al. 2010; Ito and Hayen 2011; Rossi et al. 2020); together these studies support a role for the NAcc shell in mediating the influence

of contexts on behavior. Given that spiny mice live in large groups and experience a wider range of social interactions than species that are solitary or live in small family groups, OXTRs in the NAcc shell of spiny mice may play a large role in distinguishing social context, thus facilitating context-appropriate responses to a dynamic social environment.

Interestingly, to our knowledge, OXTR binding has not been reported in the AH of rodents (but see the presence of *Oxtr* mRNA in the AH of prairie voles (Inoue et al. 2022)), however, we observed moderate OXTR binding in the AH of spiny mice, with higher binding densities in males. This finding mirrors a previous observation in spiny mice, such that they express an especially robust AH OXT neuronal population, which overlaps with the evolutionarily conserved AH AVP cell group (Kelly and Seifert 2021). Few studies have examined the role of OXT or OXTRs in the AH, however, a study that injected OXT into the “MPO-AH continuum” found a decrease in aggression in female hamsters, suggesting that AH OXT may have anti-aggressive functional properties (Harmon et al. 2002); because OXT and AVP promiscuously bind to both OXTRs and AVPR1As (Leung et al. 2009; Liu et al. 2001; Taylor et al. 2020), whether this effect is the result of OXT acting on OXTRs or AVPR1As remains unclear. An interesting consideration for future studies is to examine the role of, and potential functional sex differences in, AH OXTRs in spiny mice. It is feasible that the higher densities of OXT-immunoreactive (ir) neurons and OXTRs in the AH may serve to promote the highly prosocial and low aggressive phenotype of the spiny mice.

Species-typical group size and OXTR binding in the lateral septum

In a recent review examining OXTR binding densities in the LS across rodents that range in social group size, Olazabal and Sandberg identified that LS OXTR binding densities are lower in species that have low conspecific spacing, live in medium to large groups, and exhibit low levels of aggression compared to species that have high conspecific spacing, live in small to medium groups, and exhibit higher degrees of aggression (Olazabal and Sandberg 2020). Spiny mice fit this pattern and like colonial tuco-tucos, Norway rats, short-tailed singing mice, and guinea pigs, LS OXTR binding is relatively light in the spiny mouse brain (Beery et al. 2008; Campbell et al. 2009; Tribollet et al. 1992b, 1992a). Similar to these rodents, spiny mice also live in large communal groups and display low levels of aggression (Deacon 2009; Fricker et al. 2021; Frynta et al. 2011; Nowak 1999). Perhaps not surprisingly, this pattern does not hold true for all rodents, though. Large group-living eusocial naked mole rats and their solitary relative do not show detectable levels of OXTRs in the LS (Kalamatianos et al. 2010). Furthermore, LS OXTR binding patterns show an opposite pattern

for Estrildid finches, such that more gregarious species that live in large groups show higher septal OXTR binding densities compared to territorial, solitary finch species (Goodson et al. 2009).

While it may be somewhat informative to search for a reflection of social phenotypes within single brain regions, it is more likely that broad social phenotypes reflect diverse patterning of nonapeptide receptors, as well as other neural systems, across the brain. Indeed, species-typical group size is associated with an array of social behaviors that vary within and across species. Although some large group-living species are communal breeders and readily care for unrelated young (i.e., spiny mice), other species remain socially monogamous within large groups and care only for their own offspring or kin (i.e., zebra finches) (Tuckova et al. 2016; Zann 1996). Additionally, behaviors such as alloparental care are observed in both small and large group-living species (Kenkel et al. 2017; Tuckova et al. 2016). Along these lines, a recent meta-analysis determined that OXTR and AVPR1A binding distributions are more consistent within than across genera in rodents, suggesting that a within-genus comparison of nonapeptide receptor distributions is more appropriate than comparing across distant species (Freeman et al. 2020). This provides support for the hypothesis that distinct evolutionary selective pressures act on nonapeptide receptor patterning [in Rodentia], leading to differential receptor binding patterning across species while also urging caution when seeking to generalize nonapeptide receptor anatomy or function across species not within the same genus.

Associations with gonadal sex on OXTR and AVPR1A binding densities within the SDMN

Here we found that male spiny mice exhibited higher densities of OXTR binding in the LS, BNST, and AH compared to females. Widespread sex differences in OXTR binding have been reported for Wistar rats, again with males exhibiting greater OXTR binding densities than females (Smith et al. 2017). Similar to spiny mice, male California mice and deer mice also show more OXTR binding in the LS and BNST than females (Insel et al. 1991). However, an absence of a sex difference in LS and BNST OXTR binding has been reported for prairie voles (Bales et al. 2007). Together, this highlights that variation in nonapeptide receptor densities is not only species-specific, but also sex-specific, and that findings in one species and/or sex should not necessarily be generalized to other species and/or sexes (Dumais et al. 2013).

To explore whether nonapeptide receptor binding co-expression patterns in the SDMN differ in male and female spiny mice, we conducted Pearson correlation tests followed by hierarchical clustering analyses. Note that our sample sizes were small for this analysis and that results and

interpretations are purely exploratory in nature. We observed different correlational patterns in males and females for both OXTR and AVPR1A binding, and although only exploratory, these analyses may suggest that more strongly correlated binding densities in brain regions or clusters of brain regions could underlie sex-specific behaviors. Despite a different appearance in clustering between the sexes, examining brain region relationships for individual SDM nodes reveals some similarities between males and females. For example, OXTR binding in the Hi correlated strongly with several other brain regions in both males and females, suggesting that OXTRs in the Hi may be heavily functionally integrated in nonapeptide-mediated behavior, but not necessarily in a sex-specific manner. However, while male OXTR binding in the BNSTa correlated with binding in 3 other brain regions (LS, CPu, and Hi), BNSTa OXTR binding in females did not correlate with binding in other regions. A previous study found that BNSTa OXTR receptors promote stress-induced social avoidance in female, but not male, California mice (Duque-Wilckens et al. 2018). Together, these findings suggest that BNSTa OXTRs may coordinate activity with distinct brain regions in males and females and potentially mediate sex differences in behavior in the spiny mice. While both male and female spiny mice are socially bold and readily approach all types of conspecifics (Fricker et al. 2021), in our colony we have found that males are interactive with more individuals in a group than females and that females are more likely to exhibit aggression than males (Kelly et al. unpub obs). Given that BNSTa OXTRs in California mice promote social avoidance, similar functions may exist in female spiny mice, albeit to a different degree given the different life history and highly prosocial phenotype of the species.

The sex differences in OXTR binding reported here are consistent with our previous finding of widespread sex differences in OXT neuronal populations in the spiny mice. Males exhibited more OXT-ir neurons than females in the median preoptic nucleus, periventricular nucleus of the hypothalamus, BNST, lateral hypothalamus, and AH (Kelly and Seifert 2021). Because OXT generally promotes prosocial behaviors (Froemke and Young 2021; Goodson and Thompson 2010), future studies can seek to determine whether male spiny mice evolved a more robust OXT system to compensate for male aggression, thus promoting the prosociality needed for a communal breeding system to be successful. Indeed, sex differences in the brain do not always necessarily facilitate a sex difference in behavior, but may at times serve to prevent overt sex differences in behavior (De Vries 2004).

Although we observed no associations with gonadal sex on AVPR1A binding in brain regions throughout the SDM, we did observe different co-expression patterns in AVPR1A binding between male and female spiny mice. Not

only did we find more negative correlations in AVPR1A binding between brain regions in females than males, but the hierarchical clustering loaded 3 distinct clusters for the sexes. The most similar cluster in males and females was the MeA and VMH cluster in males and MeA, VMH, and BNSTa cluster in females. The MeA and VMH are brain regions important for the expression of defensive behaviors, aggression, and avoidance (Hashikawa et al. 2017; Hong et al. 2014; Li et al. 2017; Lin et al. 2011; Tong et al. 2020). It is intriguing that BNSTa also loaded onto this cluster in females, but not males, given that BNSTa OXTRs facilitate social avoidance in female, but not male California mice (Duque-Wilckens et al. 2020). Perhaps BNSTa AVPR1As are also involved in defensive and/or anti-social behaviors in spiny mice and play a more prominent role in facilitating those behaviors in females.

Presence of nonapeptide receptor mRNA but an absence of binding

Oxtr and *Avpr1a* mRNA was identified in 42 brain regions throughout the forebrain and midbrain of spiny mice. However, OXTR and AVPR1A binding were not observed in all sites of mRNA labeling. Within the SDM, we observed the presence of nonapeptide receptor mRNA, but absence of binding for OXTR in the PAG and for AVPR1A in the NAcc, BLA, CPu, and Hi. This is consistent with a recent characterization of *Oxtr* mRNA distributions in the prairie vole, in which a misalignment in OXT receptor mRNA expression and binding was observed in some areas of the brain (Inoue et al. 2022). Such findings suggest that nonapeptide receptors may be synthesized in a region, but the receptor protein is transported to a distal site. Axonal transport of receptor protein to neighboring brain areas can enable presynaptic signaling (Dolen et al. 2013; Yoshimura et al. 1993). Additionally, transported receptor protein may be held in latent pools within neurons until trafficked to the membrane under particular conditions (Sheng and Cai 2012). Together, this generates greater plasticity in nonapeptide system functioning and provides more avenues in which other neural systems can act upon the nonapeptide system.

A note about Manning Compound for future studies in spiny mice

In our competitive binding assay, Manning Compound did not fully block I-125 LVA binding. This suggests that either Manning Compound has a lower binding affinity in the spiny mouse compared to other species or that I-125 LVA binds to additional proteins beyond AVPR1A. Because mRNA was expressed in all regions in which receptor binding was observed, it is likely that Manning Compound has a lower binding affinity in the spiny mouse

compared to other species. This is consistent with recent findings in the Syrian hamster where Manning Compound was found to not meet criteria for selectivity for AVPR1A (Taylor et al. 2020). Therefore, caution should be taken when considering experimental designs in spiny mice that intend to use Manning Compound for pharmacological manipulation.

Conclusion

Here we characterized the distributions of OXTR and AVPR1A binding, as well as *Oxtr* and *Avpr1a* mRNA expression, throughout the basal forebrain and midbrain of male and female spiny mice. We identified an effect of sex on OXTR, but not AVPR1A, binding, in several brain regions involved in social behavior. In all regions in which a sex difference was observed, males exhibited higher binding densities compared to females. Furthermore, via clustering analyses, we identified distinct patterns of OXTR and AVPR1A binding co-expression across SDMN brain regions for males and females. These data can inform future work in spiny mice that seek to examine sex differences in nonapeptide-mediated behavior. Together, these results lay a basic foundation for future studies that seek to manipulate nonapeptide receptors or nonapeptide neural circuits (i.e., connecting peptide-producing neuronal populations with downstream receptor sites) in spiny mice, thus providing a model that is ideal for exploring nonapeptide-mediated grouping behaviors.

Author contributions JMP cryosectioned brains, conducted in situ hybridization, quantified receptor binding and mRNA expression, and wrote the manuscript. KI conducted the competitive binding assays and receptor autoradiography. KJW analyzed the data. AWS provided feedback on and edited the manuscript. LJY designed the study, provided feedback on and edited the manuscript, and acquired funding. AMK designed the study, analyzed the data, wrote the manuscript, and acquired funding.

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Data availability Data are available upon request from the corresponding author.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

Ethics approval All procedures were approved by the Institutional Animal Care and Use Committee of Emory University.

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